

SILANES

Definition

Silanes are a large kinds of charge-neutral compounds with the formula SiR_4 . The R substituents can be an organic or in-organic group or groups. They may also contain one, two, three or four Si---H bonds and are called hydrosilanes. Thus, hydrosilanes are compounds of silicon and hydrogen. Common examples of hydrosilanes are; triethylsilane ($\text{HSi}(\text{C}_2\text{H}_5)_3$), triethoxysilane ($\text{HSi}(\text{OC}_2\text{H}_5)_3$) and dodecamethylcyclohexasilane ($((\text{CH}_3)_2\text{Si})_6$). Other types of silanes are Polysilanes, chlorosilanes and the organosilanes.

Polysilanes are organosilicon compounds with the formula $(\text{R}_2\text{Si})_n$ and those of interest here are the organic derivatives such as polydimethylsilane ($\text{Si}(\text{CH}_3)_2$).

Chlorosilanes have Si---Cl bonds and prominent examples are: trichlorosilane (SiHCl_3) and tetrachlorosilane (SiCl_4).

While organosilanes are a class of charge-neutral organosilicon compounds, e.g. tetramethylsilane $\text{Si}(\text{CH}_3)_4$.

Silicon can form a series of hydrides with the general formula $\text{Si}_n\text{H}_{2n+2}$. The hydrides so formed could be such as SiH_4 , Si_2H_6 , Si_3H_8 ,, the compounds either have the Si---H or Si---Si single bonds. The possession of empty valence shell d-orbitals by Si gives it the ability to form Silanes.

A silane is therefore, chemically defined as a binary compound of silicon and hydrogen obtained by treating ionic silicide e.g. magnesium silicide with a dilute (aqueous) acid e.g. hydrochloric acid. For instance, the preparation of the simplest silane SiH_4 can be industrially/commercially by the action of dilute HCl with Mg_2Si .

First case: $\text{Mg}_2\text{Si} + \text{H}^+ \rightarrow 2\text{Mg} + \text{SiH}_4 + \text{H}_3\text{Si}^+ + \dots\dots\dots \text{SiH}_2 + \dots\dots\dots$ This reaction leads to the formation of mixed hydrides that are industrially separated by fractional distillation process, while the individual isomers are then separated by vapour-phase chromatography.

And also by reducing silicon tetrachloride with a lithium-aluminium hydride, the first member of the series tetrahydrosilane (SiH_4) can be produced.

Second case: $\text{SiCl}_4 + \text{LiAlH}_4 \rightarrow \text{SiH}_4 + \text{LiCl} + \text{AlCl}_3$.

Physical properties of Silanes

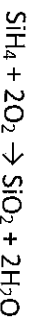
1. Silanes are gases or volatile liquids with increasing boiling points as molecular weights increases.
2. Silanes are very stable to heat. But their stability decreases as chain length increases.
3. The first member SiH_4 is a colourless gas and thermally stable at room temperature. The second member is also gaseous at room temperature. While the 3rd and 4th are liquids under same conditions.
4. Silanes experience inter-molecular Vander Waals forces which gives rise to their higher boiling and melting points.

Table 1: Common silanes and their physical properties.

Name of Silane	Formula	B.P. °C	M.P. °C	Form
mono silane	SiH_4	-12.0	-185	Gas
disilane	Si_2H_6	-14.0	-132	Gas
trisilane	Si_3H_8	53.0	-117	Liquid
tetrasilane	Si_4H_{10}	100.0	-84	Liquid

Chemical properties of Silanes

1. Silanes reacts violently with air to produce silicon dioxide and water:



This reaction is analogous to the combustion of hydrocarbons to yield CO_2 and H_2O .

2. Silanes react stepwisely with hydrogen halides to produce halogenated silanes. For instance, the primary silane reacts with HBr , to yield all possible combinations of brominated silane from SiH_3Br to SiBr_4 .
3. Silanes are also slightly hydrolysed in water, but rapidly in the presence of alkalis producing hydrogen gas and silicate ions:



In contrast, hydrocarbons on the other hand are inert to hydroxides.

The most important reaction of silanes from a commercial viewpoint, is their reactions with alkenes to yield organo halosilanes which are used to produce silicone

polymers through hydrolysis and further polymerization reactions. Example of such reaction involving dichlorosilane with propene is as follows:



The Vander waals forces in silanes are relatively strong as evidenced by the following two facts:

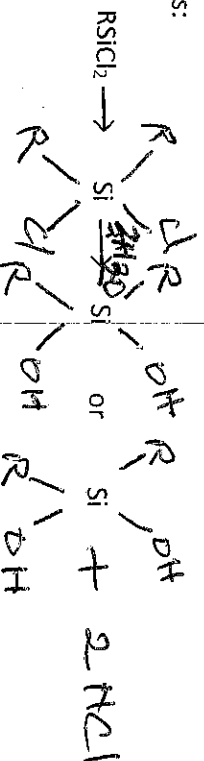
1. The number of electrons surrounding the molecule, which increases with the increasing molecular weights of the silanes.
2. The surface area of the molecule.

Straight chain silanes exhibit higher boiling points than their branch chain counterparts due to the greater surface area in contact, thus the greater Vander waals forces between adjacent molecules. Conversely, cyclosilanes tend to have higher boiling points than linear silanes due to the locked conformations of the molecules, which gives rise to a plane of intermolecular contact.

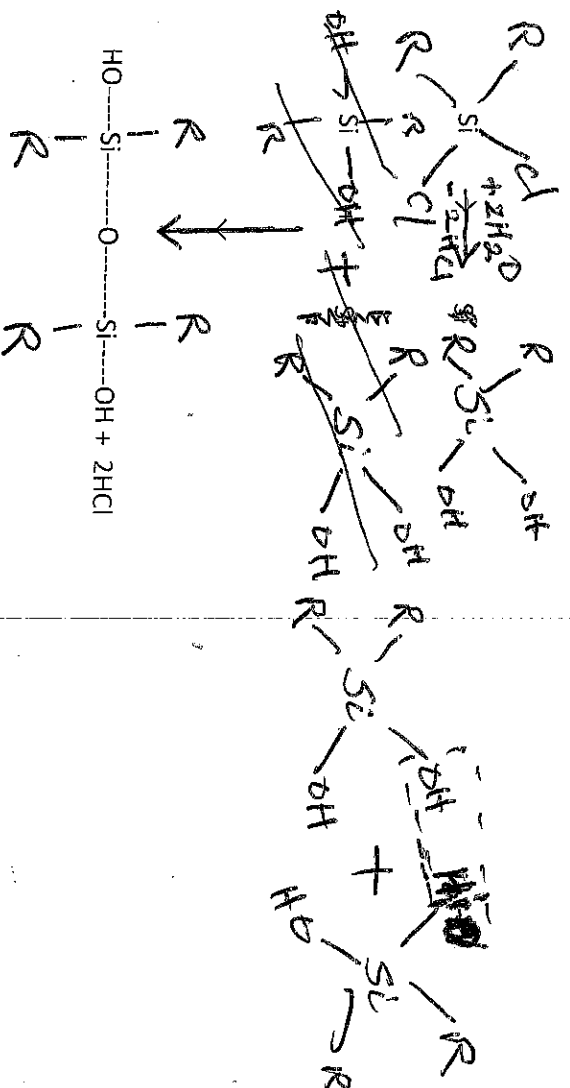
Silanes are non-conductors of electricity, and are not highly polarised by an electric field. That is why they do not form hydrogen bonds and hence insoluble in polar solvents like water. The rate of decomposition of silanes increases with increasing number of silicon atoms in the molecule. Silanes also have the ability to incorporate the same functional groups as their alkane counterparts. e.g. the $-\text{OH}$, to make silanol as in alcohol or methanol and a halogen to make silicon halide analogous to alkyl halide. *(note the two reactions)*.

Hydrolysis of alkyl chlorosilanes

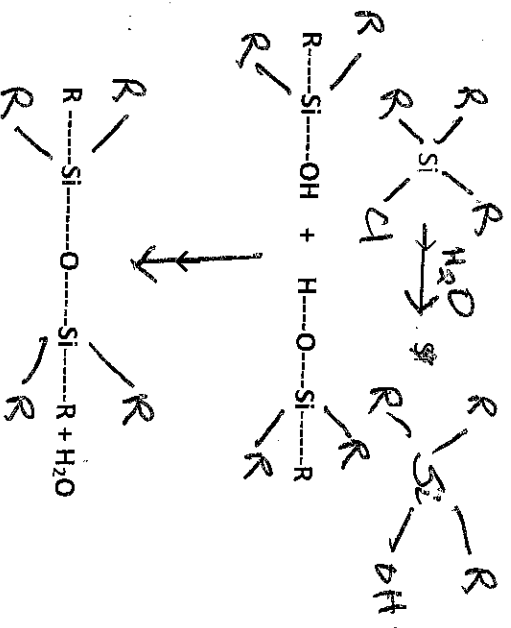
The hydrolysis of alkyl substituted chlorosilanes is a very important reaction that leads to the production of a very useful polymer called the silicone-polymers. This reaction proceeds by the action of water on the alkyl substituted chlorosilane where the chlorine atoms are substituted by the OH of the water forming a $\text{Si}-\text{O}-\text{Si}$ linkages and the formation of HCl. It is generally represented as follows:



For example, in the hydrolysis of alkyl substituted chlorosilane shown below:



Similarly, the hydrolysis of a trialkylchlorosilane can be shown as follows:



Hydrolysis of typical (named) alkyl chlorosilanes

Alkyl chlorosilanes undergo hydrolysis to produce linear, cyclic and cross-linked silicone polymers upon polymerization of the hydroxyl organosilane intermediate formed. For example, the hydrolysis of trialkyl monochlorosilane as shown below:

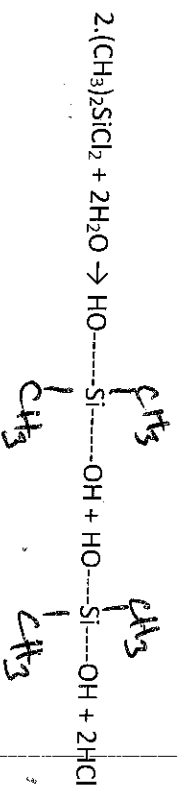


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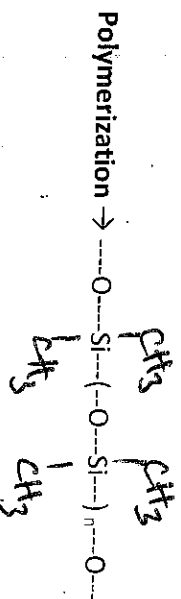
(Hydroxy organosilane intermediate). This compound will then undergo polymerization reaction yielding the linear silicone polymer.

Polymerization $\rightarrow$ $(\text{CH}_3)_3\text{Si}-\text{O}-\text{Si}(\text{CH}_3)_3$ (hexamethylsiloxane polymer)

Another instance is the hydrolysis of a dialkylchlorosilane (dimethylchlorosilane) shown as follows:



and these two will upon polymerization reaction yield the linear siloxane polymer.



Uses / application of Silanes

There is a wide industrial and medical application of silanes, especially the primary silane SiH_4 .

For example, silanes are used as coupling agents to adhere glass fibers to a polymer matrix, stabilizing the composite material.

They are used to couple a bio-inert layer on titanium implants due to the presence of organic and inorganic attachments.

Note: (Implants are medical devices manufactured to replace a missing biological structure, support a biological structure or even enhance a biological structure).

Silanes are used as water repellents due to their hydrophobicity. They are also used for masonry protections and for the control of graffiti.

The primary silane being a reducing agent, is used for the removal of corrosion in inaccessible plants. e.g. ~~in~~ the inside of pipes in nuclear reactors.